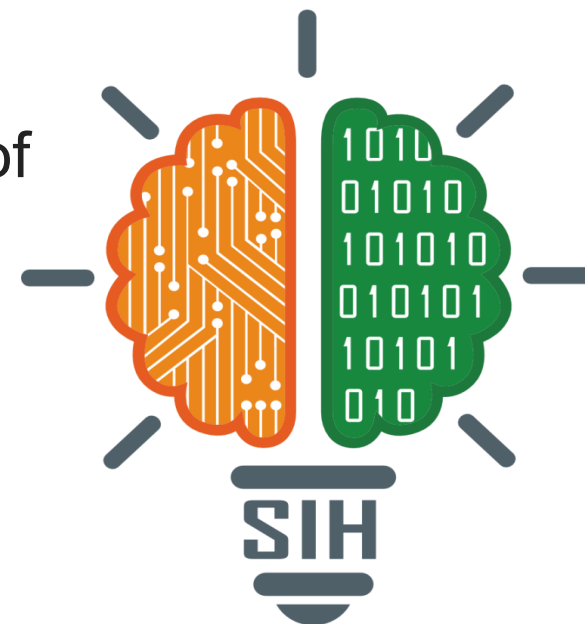


## FOG-SAFE: Offline Cooperative Collision Warning for HEMM Operations in Severe Fog

- **Problem Statement ID** - SIH26007
- **Problem Statement Title**- Safe and Efficient Operation of Mine Vehicles in Fog and Low-Visibility Conditions in Open Cast Iron Ore Mines
- **Theme** - Smart Automation
- **PS Category** - Hardware
- **Team ID** - 152900
- **Team Name** - Aegis Vision



# FOG-SAFE: Offline Cooperative Collision Warning for HEMM Operations in Severe Fog

## Problem:

- Severe fog hides vehicles, workers and obstacles.
- HEMM blind zones and blind curves delay hazard recognition.
- Remote mine areas may lack reliable internet connectivity.

## Solution Overview:

An offline cooperative driver-assistance system that allows mine vehicles and workers to sense, assess and share hazards beyond the driver's visible range.

## Proposed Solution:

- **360° sensing:** mmWave radar, thermal camera, Lidar and visibility sensor
- **Local intelligence:** calculates hazard direction and collision risk
- **Vehicle context:** GNSS/IMU provides position, heading and movement
- **Direct communication:** offline V2V and V2W warning exchange
- **Directional alerts:** cabin display, audio and haptic safety jacket
- **Retrofit design:** installable on existing mine vehicles.

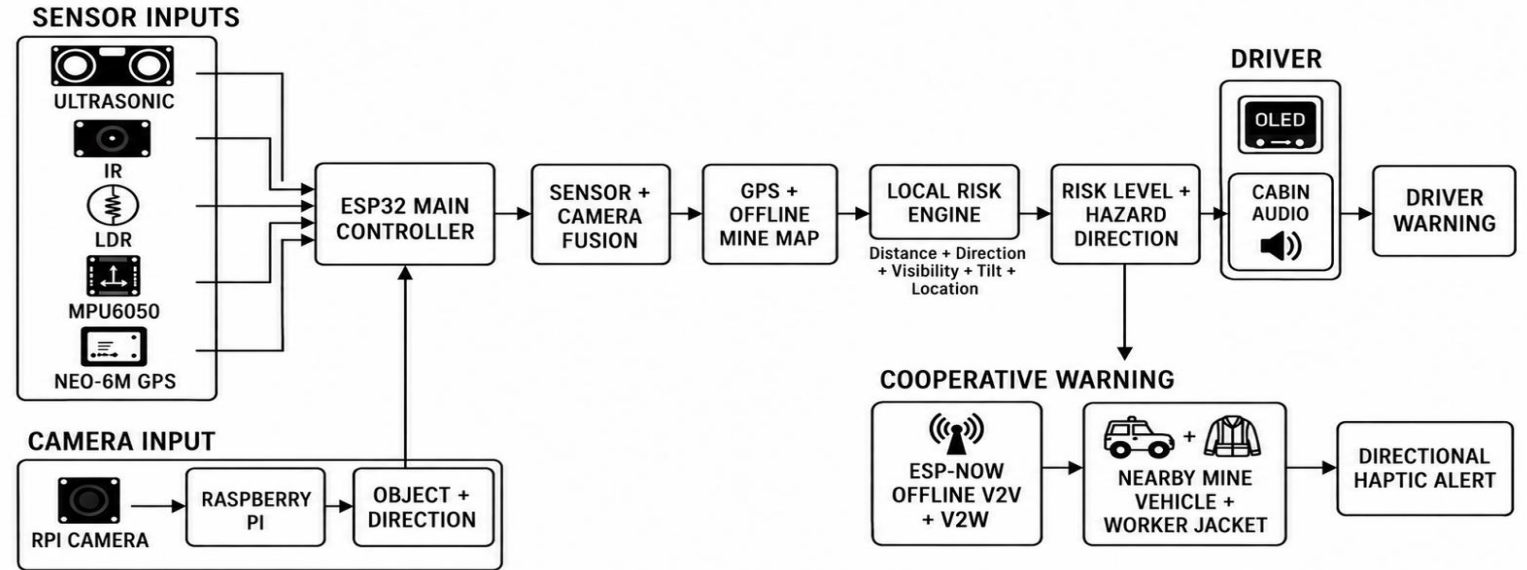


## Innovation/Uniqueness:

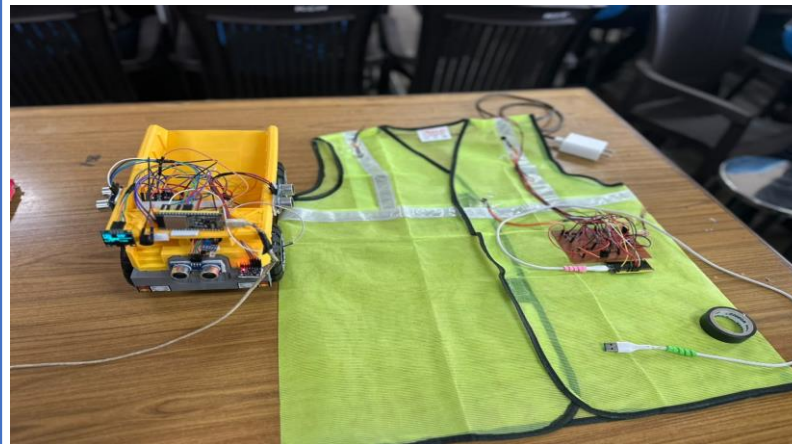
- Beyond-line-of-sight hazard awareness
- Vehicles and workers operate as cooperative safety nodes
- Offline-first operation without cloud dependency
- Directional warnings show where the danger is.
- Driver-assistance only; operator retains vehicle control
- Scalable from one high-risk zone to the mine fleet

## Hardware & Software:

- **ESP32:** Sensor fusion, risk calculation and alert control
- **Raspberry Pi + Camera:** Computer-vision detection of vehicles, workers and obstacles
- **Ultrasonic + IR Sensors:** Obstacle detection and distance measurement
- **LDR:** Low-visibility detection
- **MPU6050 + GPS:** Vehicle orientation and position tracking
- **ESP-NOW:** Offline V2V and V2W warning communication
- **OLED, Cabin Audio + Haptic Jacket:** Directional driver and worker alerts
- **Embedded C++ +, Python:** Sensor control and computer-vision processing
- **Local Risk Engine + Offline Map:** Hazard assessment and high-risk-zone identification



## Prototype:



Youtube Link:

<https://youtube.com/watch?v=vNG1HZtww0I&feature=shared>

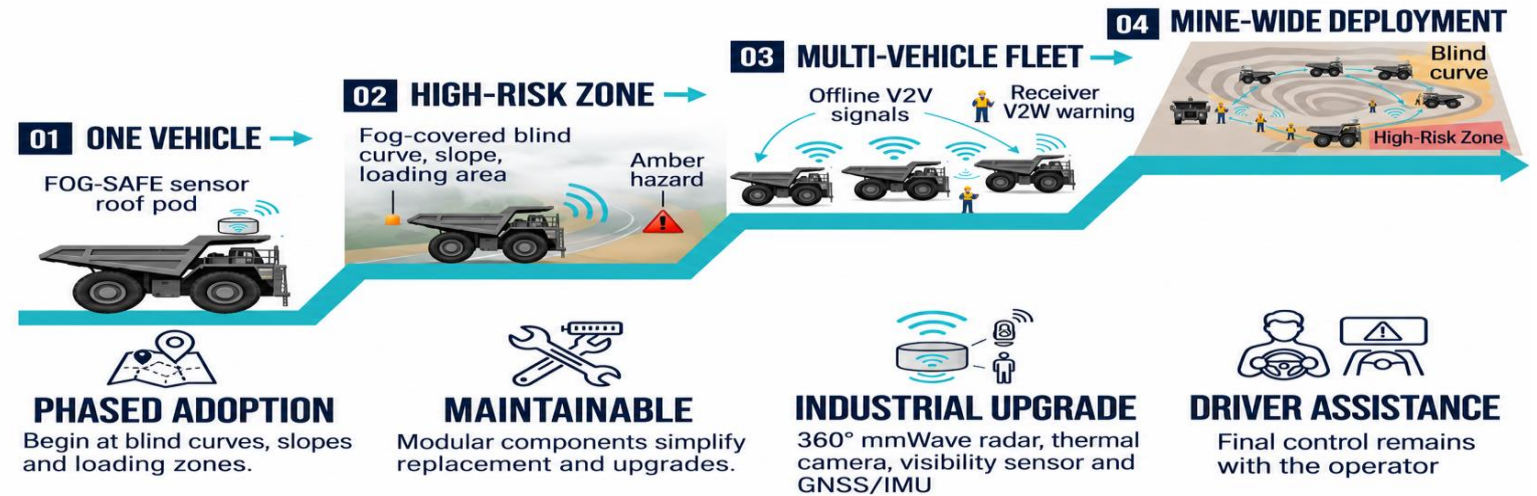
Github Link:

<https://github.com/rnasvitha/Fog-Safe.git>

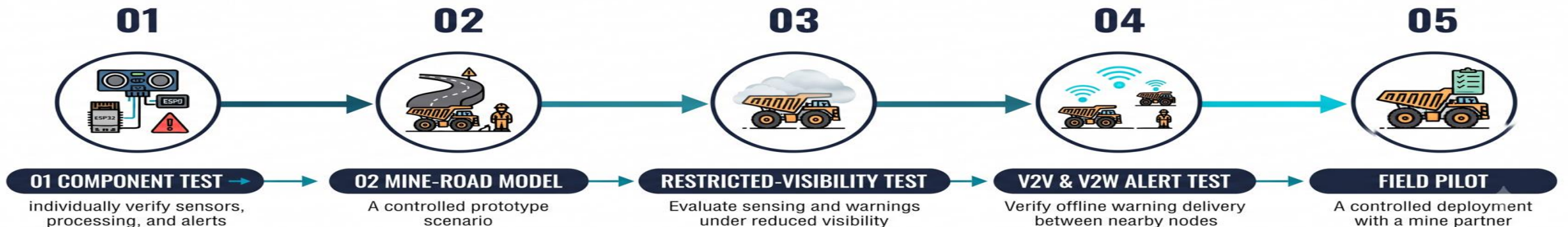
## Technical Feasibility:

- **Available Hardware:** Readily available prototype components
- **Edge Processing:** ESP32 risk logic + Pi camera processing
- **Offline Operation:** ESP-NOW communication without internet
- **Retrofit Design:** Modular installation on existing vehicles
- **Industrial Upgrade:** Radar, thermal, visibility, Lidar and GNSS/IMU

## Deployment Viability:



## Validation Path:



## Potential impacts:

### Mine Workers

- Early hazard warnings
- Directional vibration alerts
- Better awareness in dense fog

### Vehicle Drivers

- Hazard information beyond visual detection
- Alerts for workers, vehicles & obstacles
- Faster reaction to critical situations

### Mine Owners

- Scalable multi-vehicle safety system
- Modular and upgradeable architecture
- Complements existing safety infrastructure

- <1 sec hazard-alert generation
- 360° surrounding awareness
- 3–5 m low-visibility scenario
- 4-direction haptic guidance

## Benefits of the solution:

### SOCIAL

- Improves worker awareness and safety
- Reduces dependence on visibility alone
- Provides simple directional alerts

### ENVIRONMENTAL

- Low-power wireless communication
- Reduces additional wired infrastructure
- Supports efficient vehicle operation

### OPERATIONAL

- Vehicle → Vehicle communication
- Vehicle → Worker communication
- Real-time vibration + audio alerts

### ECONOMIC

- Low-cost prototype architecture
- Supports modular deployment
- ₹1–₹3 lakh/vehicle projected industrial retrofit
- 30–50% lower implementation cost

## Business model:

### Target Customers

- Open-cast mining companies
- Mining contractors & fleet operators
- Heavy-vehicle safety departments
- Mine equipment / automation integrators

### Deployment Model

- Pilot Deployment → Mine-wide Deployment → Multi-site Expansion

### Key Partnerships

- Mining companies
- Mining equipment manufacturers
- Industrial automation companies
- Safety-system integrators

### Revenue Model

- Vehicle safety hardware
- Smart worker jacket
- Annual maintenance & support
- Industrial retrofit: ₹1–₹3 lakh/vehicle
- Annual maintenance: 8–12% of system cost

## 01 — DGMS: Metalliferous Mines Regulations, 1961

*“Provides the Indian regulatory foundation for safe machinery operation, haulage, lighting and worker protection in metalliferous mines.”*

<https://www.dgms.gov.in/>

## 02 — ISO 5006:2017: Operator’s Field of View

*“Specifies methods for evaluating operator visibility and blind areas around earth-moving machinery.”*

<https://www.iso.org/standard/45609.html>

## 03 — Bijelic et al., CVPR 2020: Seeing Through Fog Without Seeing Fog

*“Demonstrates how multimodal sensor fusion improves object detection under fog and other adverse-weather conditions.”*

[https://openaccess.thecvf.com/content\\_CVPR\\_2020/html/Bijelic\\_Seeing\\_Through\\_Fog\\_Without\\_Seeing\\_Fog\\_Deep\\_Multimodal\\_Sensor\\_Fusion\\_CVPR\\_2020\\_paper.html](https://openaccess.thecvf.com/content_CVPR_2020/html/Bijelic_Seeing_Through_Fog_Without_Seeing_Fog_Deep_Multimodal_Sensor_Fusion_CVPR_2020_paper.html)

## 04 — Gelbal et al., 2023: Cooperative Collision Avoidance

*“Presents collision-risk assessment using information exchanged directly between nearby connected vehicles.”*

<https://arxiv.org/abs/2306.01889>

## 05 — S. Hasirlioglu et al., 2017: Foggy Weather Detection for Automotive Applications

*“Investigates sensor-based detection and perception challenges in foggy weather conditions for automotive applications.”*

<https://ieeexplore.ieee.org/document/7995808>